

Landslide Prediction Approaches using AI Techniques - A Literature Review

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Abstract— Landslides are the natural disasters often result into severe socio-economic and environmental consequences. Due to which, early warning systems with precise landslide predictions are imperative to be introduced. In this paper, machine learning as well as deep learning-based landslide predictions techniques are critically and comparatively reviewed, introduced by existing authors. In this review, different techniques have been identified such as ‘Convolutional Neural Networks’ (CNNs), ‘Recurrent Neural Networks’ (RNNs), ‘Transformer models, Long Short-Term Memory’ (LSTM) networks, and hybrid models, for landslide prediction. After finding literature gaps, a new methodology is proposed for landslide prediction in a better manner. The proposed approach is a novel approach composed of transformed, autoencoder, and logistic regression. Unlike existing techniques, the model promises to conduct real-world landslide prediction with higher accuracy and scalability.

Keywords—Landslide Prediction; Deep Learning; Machine Learning;

I. INTRODUCTION

Landslides are one of the most destructive natural disasters in the world, responsible for 17% of all deaths worldwide. Andes, the Alps, the Rocky Mountains, and the Asian highlands, are highly prone to the landslide occurrences [1]. Majorly, the landslides are occurred due to earthquakes, volcanic eruptions, and rains, and observed in US, China, India, Nepal, and Italy, frequently. Also, the changing patterns of precipitation, and extreme weather conditions, responsible for frequency and size of landslides [2]. Annually, \$20 billion lives are lost worldwide due to landslides, with agricultural, traffic system, and infrastructure damages [3].

India's diverse terrain makes it especially vulnerable; the Geological Survey of India has classified more than 15% of its land area as landslide-prone [4]. Most impacted are the northeastern states, the Western Ghats, and the Himalayan region. Approximately 1,000 landslides are reported annually in the Himalayas alone, and they are frequently made worse by the monsoon season [5]. More than 400 people were killed and hundreds were displaced in Himachal Pradesh in 2023 as a result of more than 300 recorded landslides brought on by intense monsoon rainfall [6]. In addition to taking lives, these occurrences have a negative influence on the economy, access to healthcare, education, and food security.

Real-time forecasting and spatial coverage are two limitations of traditional landslide prediction techniques, which are based on field surveys, geological mapping, and

historical data. However, the capacity of ‘machine learning’ (ML) and ‘deep learning’ (DL) models to handle huge, diverse datasets has led to their considerable popularity. In regional landslide susceptibility mapping, machine learning methods like ‘Random Forests’ (RF) and ‘Support Vector Machines’ (SVM) have shown above 80–90% accuracy [7]. Patterns in satellite images and LiDAR data can be effectively identified by Deep Learning models, especially ‘Convolutional Neural Networks’ (CNNs). In order to increase the accuracy of predictions in both space and time, hybrid systems that combine ML and DL techniques are being developed more and more [8].

A thorough analysis of current developments in ML, DL, and hybrid approaches for landslide prediction is provided in this review paper. Key datasets, methods, performance indicators, and application case studies are highlighted. In order to improve early warning systems and decision-making in landslide-prone areas, the assessment also highlights current gaps and difficulties and suggests future research avenues.

II. RESEARCH METHODOLOGY: QUALITATIVE METHODOLOGY

This review adopts a qualitative methodology to synthesize existing research on landslide prediction, focusing on the application of ‘machine learning’ (ML), ‘deep learning’ (DL), and hybrid models.

A. Review Process

The review process involved the systematic identification, selection, and evaluation of scholarly articles published in peer-reviewed journals and conferences. Sources were retrieved from reputable databases including IEEE Xplore, ScienceDirect, SpringerLink, Scopus, and Google Scholar, using search terms such as “landslide prediction,” “machine learning,” “deep learning,” “GIS-based modeling,” and “remote sensing.”

Studies were included based on relevance, clarity of methods, and contributions to landslide prediction techniques. Emphasis was placed on the modeling approaches, input data types (e.g., topographic, hydrologic, geological), evaluation metrics, and application regions.

Articles were grouped and compared by modeling category to assess trends, strengths, and limitations. This approach allows for a thematic understanding of current progress and challenges, providing a foundation for identifying research gaps and future directions.

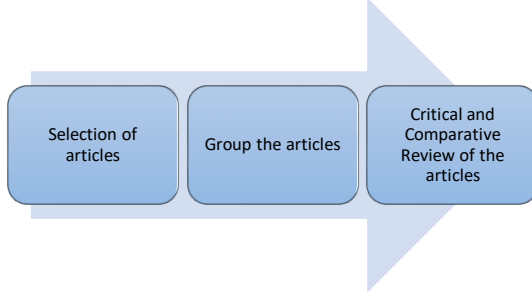


Figure 1: Review Process

III. RELATED WORK

Landslide prediction has evolved from traditional statistical methods to modern data-driven techniques. Earlier models such as logistic regression, frequency ratio, and analytical hierarchy process relied on historical data and expert knowledge but often struggled with nonlinearity and complex terrain interactions. Machine Learning (ML) methods like Random Forests, Support Vector Machines, and Decision Trees offer higher accuracy by capturing complex patterns among features such as slope, rainfall, soil type, and land use. More recently, ‘Deep Learning’ (DL) models, especially ‘Convolutional Neural Networks’ (CNNs), have shown strong performance in analyzing satellite imagery and ‘Digital Elevation Models’ (DEMs) to detect spatial patterns. ‘Recurrent Neural Networks’ (RNNs) and LSTMs are applied for time-series rainfall-triggered landslides. Hybrid approaches combining ML, DL, GIS, and remote sensing further enhance prediction performance. Despite progress, challenges remain in data availability, generalization across regions, and model interpretability, which continue to guide ongoing research in landslide prediction.

A. LANDSLIDE PREDICTION USING MACHINE LEARNING MODELS

Wu, B., Shi, Z., Zheng, H. et al. [1] contribute an integrated approach using multi-source data—morphological, geological, hydrological, and land-use information from northwest Oregon—to develop and train four machine learning models (Decision Trees, Random Forest, Adaboost, and Gradient Tree Boosting) for landslide susceptibility prediction. Models trained on imbalanced datasets demonstrated better classification performance, favoring negative classifications at lower susceptibility levels, whereas balanced datasets achieved higher positive predictions (landslides) in high-susceptibility areas with accuracy of 91.2%. By employing the Shapley Additive Explanations (SHAP) method, the study identified five key factors influencing landslide susceptibility: proximity to roads, slope roughness, geological age, terrain roughness, and elevation. Analysis of False Negatives (FN) and False Positives (FP) emphasized that FN errors could lead to severe consequences, including loss of life, while FP errors,

though less critical, may stem from data limitations or impending landslides. The study recommends using balanced datasets to minimize FN and optimize landslide capture in high-susceptibility regions. These insights highlight the role of dataset balance in model sensitivity to landslide risks and suggest approaches for refining data sampling to improve prediction accuracy.

Then, Nanda, A.M., Lone, F.A. & Ahmed, P. investigate machine learning models— ‘Random Forest’ (RF) and ‘Logistic Regression’ (LR) in predicting rainfall-induced landslides along the Bandipora to Gurez highway in Jammu & Kashmir, India, using metrics such as ROC-AUC and ‘false-negative rate’ (FNR). Amongst them, ‘Logistic Regression’ (LR) showed a lower error rate, with an 8.48%. Thus, LR proved more effective and is recommended for landslide prediction due to its acceptable FNR range. The study supports broader implementation of machine learning models in early landslide warning systems, emphasizing the value of antecedent rainfall data for improved accuracy [2]. In comparison, Hailu et al utilizes Logistic Regression along with Frequency Ratio models, for the landslide prediction, and AUROC values ranging from 80%–90 and 70%–80, respectively [23]

Later Javidan, N., Kavian, A., Conoscenti, C. et al aim to create multi-hazard probability maps for floods, landslides, and gully erosion in the Gorganrood Watershed, utilizing machine learning models such as Random Forest (RF), ‘Support Vector Machine’ (SVM), ‘Boosted Regression Tree’ (BRT), and ‘Multivariate Adaptive Regression Splines’ (MARS). Among the models, MARS showed the highest predictive performance across hazards with an AUC of 99.1%. For individual hazards, RF performed excellently for floods and well for landslides. Variable importance analysis highlighted drainage density (89.4%) for floods, digital elevation model (30.5%) for landslides, and rainfall (41.7%) for gully erosion as critical factors. These multi-hazard maps serve as essential tools for natural resource conservation, environmental protection, and informed land use planning in areas vulnerable to multiple hazards [3].

Thereafter Madhu, D., Nithya, G.K., Sreekala, S. et al. focus on landslide susceptibility mapping in Kerala’s Idukki district, India. A comprehensive database of landslide events was compiled from literature, ‘Geological Survey of India’ (GSI) reports, and news sources. ‘Geographic Information System’ (GIS) mapping was employed to represent landslide causative factors identified in previous studies. The machine learning models applied include Decision Tree, Logistic Regression, K-Nearest Neighbor, Gaussian Naive Bayes, and Support Vector Machine (with linear, polynomial, and Gaussian kernels), alongside ensemble methods like Random Forest and AdaBoost. Model performance was evaluated using statistical measures like G-Mean, F1 score, and ROC-AUC. Findings indicate that machine learning models effectively generate landslide susceptibility maps, with all algorithms producing reliable results, supporting the effectiveness of these techniques in identifying landslide-prone areas [4].

Sr. No.	Author	Approach	Algorithm Used	Accuracy Achieved
[1]	Wu, B., Shi, Z., Zheng, H. <i>et al.</i>	Machine Learning	Decision Trees, Random Forest, Adaboost, and Gradient Tree Boosting	AUC of 91.2% with Random Forest
[2]	Nanda, A.M., Lone, F.A. & Ahmed, P.	Machine Learning	Random Forest (RF) and Logistic Regression (LR)	AUC of 92% with Random Forest
[3]	Javidan, N., Kaviani, A., Conoscenti, C. <i>et al.</i>	Machine Learning	Random Forest (RF), Support Vector Machine (SVM), Boosted Regression Tree (BRT), and Multivariate Adaptive Regression Splines (MARS)	AUC of 99.1% using Multivariate Adaptive Regression Splines (MARS)
[4]	Madhu, D., Nithya, G.K., Sreekala, S. <i>et al.</i>	Machine Learning	Logistic Regression	AUC of 91%

B. LANDSLIDE PREDICTION USING DEEP LEARNING MODELS

Li, J., Fan, C., Zhao, K. *et al.* [11] explored that landslide displacement prediction using ‘interferometric synthetic aperture radar’ (InSAR) data faces two challenges: InSAR’s single-dimensional deformation data only reflects movement along the satellite’s ‘line of sight’ (LOS) and doesn’t capture displacement along the slope. Additionally, using a single model limits prediction accuracy as it struggles to address both long-term and localized changes in displacement.

To overcome these limitations, Li, J., Fan, C., Zhao, K. *et al.* proposes a hybrid LSTM-TCN (Long Short-Term Memory-Temporal Convolutional Network) model. First, InSAR-derived surface deformation data is reprocessed, then LSTM-TCN model is applied, combining to address both extensive temporal trends and short-term

variations. The model achieved R^2 (0.75), RMSE (0.43 cm), and MAE (0.36 cm). These results indicate that the model captures complex patterns more effectively, improves prediction accuracy for landslide displacement, enhancing landslide monitoring, and offering a robust method for real-world applications [11].

Then, Li, B., Li, C., Liu, Y. *et al.* introduces a distributed deep learning (DDL) approach, that is collaborative Feature Neural Network (DDL-CFNN) model, allowing multiple landslide-monitoring institutions by sharing model features rather than raw data. It permits indirect dataset expansion, improves model performance without compromising data privacy. The DDL-CFNN model achieves 62%–76% lower prediction errors and offers a promising framework for advancing landslide forecasting, and supports geohazard management efforts [5]. Later, Wang, Y., Zhou, C., Cao, Y. *et al.* emphasizes ‘Landslide Susceptibility Mapping’ (LSM) by recognizing the diversity in landslide mechanisms. For susceptibility assessment, CNN model is proposed and evaluated via ‘Receiver Operating Characteristic’ (ROC) and statistical analyses, outperformed its counterparts, suggesting that incorporating landslide-specific characteristics enhances prediction accuracy up to 92% [6].

In comparison, Saha, S., Barman, A., Saha, A. *et al.* focuses on landslide vulnerability in the Nilgiri district of Tamil Nadu, India, using advanced ‘deep learning’ (DL) models. A DLNN, with an AUC of 89.07%, offers superior accuracy, reveals that 43.31% and 37.72% of the district fall under high to very high vulnerability categories. This framework can assist planners in making data-driven decisions for improved landslide risk management in vulnerable areas [7].

Sr. No.	Author	Approach	Algorithm Used	Accuracy Achieved
[5]	Li, B., Li, C., Liu, Y. <i>et al.</i>	Deep Learning	‘DDL-based Collaborative Feature Neural Network’ (DDL-CFNN)	76% lesser prediction errors
[11]	Li, J., Fan, C., Zhao, K. <i>et al.</i>	Deep Learning	LSTM-TCN	AUC of 97.6%
[6]	Wang, Y., Zhou, C., Cao, Y. <i>et al.</i>	Deep Learning	CNN	AUC of 92%
[7]	Saha, S., Barman, A., Saha, A. <i>et al.</i>	Deep Learning	DLNN	AUC of 89.07%

development of targeted preventive strategies with 89% AUC.

C. LANDSLIDE PREDICTION USING COMPUTER VISION

Over the past decade, ‘remote sensing’ (RS) image-based landslide forecasting has gained significant attention, as early detection of landslide-prone areas can save lives and reduce economic loss. However, this method faces challenges since traditional image processing is insufficient for analyzing RS images.

Thus, Varangaonkar et. al [12] introduces a new landslide prediction method combined with a severity analysis model, leveraging real-time hyperspectral RS images. The model consists of pre-processing, dynamic segmentation, hybrid feature extraction, landslide prediction, and severity detection phases. In the pre-processing stage, geometric correction and ‘Normal Difference Vegetation Index’ (NDVI) are applied to remove irrelevant features like urban areas, water, and vegetation. Additional steps include atmospheric corrections, geometric adjustments, and denoising with 2D median filtering. Dynamic segmentation is then applied to localize the ‘Region of Interest’ (ROI), from which hybrid features are extracted to capture spatial and temporal variations. After normalizing each feature vector, ‘Artificial Neural Network’ (ANN) and ‘Support Vector Machine’ (SVM) models are trained to predict landslides. Upon detecting a landslide, the model assesses its severity. Real-time RS images from Goa and Maharashtra were used for performance testing, with results demonstrating the proposed model’s effectiveness with 87.6% accuracy.

In comparison to Varangaonkar et. al [12], Bhuvaneshwari et. al [13] uses the Landslide4sense dataset, comprising 3,799 training samples and 245 testing samples. Image patches come from the Sentinel-2 sensor, with slope and ‘Digital Elevation Model’ (DEM) data from the ALOS PALSAR sensor. Data were collected from Kodagu, Iburi, Taiwan, and Gorkha. To predict landslides, Deep Learning (DL) models like ResNet18, U-Net, and VGG16 were employed. Through comparative analysis based on evaluation metrics including loss, precision, recall, F1 score, and accuracy, ResNet18 emerged as the most effective model for landslide detection with [13].

In contrast to it, Qiu, Q., Xie, Z., Zhang, D. et al. [14] introduce an automated geological hazard knowledge graph, designed to support safety risk analysis by leveraging computer vision and geoscience ontology. The system identifies geological hazards from input images while adhering to safety protocols, even under changing conditions. The knowledge graph’s ontology schema for geological disasters is created using a top-down approach, enabling knowledge sharing and semantic interoperability through logical expressions. Computer vision methods detect entities and attributes from image data, categorizing object types and attributes for storage in the Neo4j database. A reasoning model within Neo4j establishes nodes, relationships, and properties for effective hazard modeling. Geological hazards are identified by querying the Neo4j database, which improves hazard detection and informs preventive measures. Results confirm the model’s effectiveness in pinpointing potential hazards, aiding in the

Sr. No.	Author	Approach	Algorithm Used	Accuracy Achieved
[12]	Li, J., Fan, C., Zhao, K. <i>et al</i>	Computer Vision	Feature vector, Artificial Neural Network (ANN) and Support Vector Machine (SVM) models	AUC of 87.6%
[13]	Wang, Y., Zhou, C., Cao, Y. <i>et al</i> .	Computer Vision	ResNet18, U-Net, and VGG16	AUC of 91% with ResNet
[14]	Qiu, Q., Xie, Z., Zhang, D. <i>et al</i> .	Computer Vision	automated geological hazard knowledge graph	AUC of 89%

D. OTHER HYBRID APPROACHES FOR LANDSLIDE PREDICTION

Şener, A., Ergen, B. [8] introduces a novel semantic segmentation model for early landslide detection using remote sensing imagery. The model includes encoder-decoder architecture, combines local and contextual data, enables precise detection of landslide areas with fewer parameters. It makes the model, computationally efficient choice for landslide segmentation. Also, the model achieved a remarkable accuracy of 97.60% and a mean Intersection over Union (mIoU) of 73.65%. The high accuracy and efficiency underscore the model’s potential for providing a valuable tool for early intervention and effective disaster response [8].

In contrast, Chen, C., Fan, L. [9] examine the choice of contributing factors, and their impacts on accuracy of landslide predictions in machine learning (ML) and deep learning (DL) models. ‘Information Gain Ratio’ (IGR), ‘Particle Swarm Optimization’ (PSO), ‘Least Absolute Shrinkage and Selection Operator’ (LASSO), ‘Recursive Feature Elimination’ (RFE), and ‘Harris Hawk Optimization’ (HHO) are used to evaluate the factor selection methods. Additionally, autoencoder-based an exhaustive approach was applied, to assess all possible factor combinations as benchmarks. Results reveal that ML models benefit from selecting important contributing factors, which enhance predictive accuracy. Interestingly, IGR and RFE-based selections decreased the predictive accuracy for DL models, while autoencoder-based

selections improved it by AUC of 93%. The study concludes that the choice of selection methods plays a more significant role in improving landslide prediction accuracy than the selection of specific contributing factors [9].

Wu, Zhang, Y., Yang, L. et al. [10] investigates critical landslide factors and examines how varying grid resolutions affect ‘landslide susceptibility mapping’ (LSM) on the Sino-Pakistani Karakoram Highway. Using InSAR data, 21 influencing factors were identified, and feature selection methods such as ‘Spearman Correlation Coefficient’ (SCC), ‘Random Forest Feature Selection’ (RFFS), and a combined SCC-RFFS, were applied across six grid resolutions (10 m to 3000 m). Machine learning models built with both original and selected factors were then compared. Results show that the relief degree of land surface (RDLS), slope, and rainfall are the most influential factors on landslide occurrence and remain consistent across resolutions, while secondary factors like ‘Topographic Wetness Index’ (TWI) vary. The SCC-RFFS-RF model achieved the highest accuracy at 30 m resolution. This approach supports efficient land resource management and enhances safety and cost-effectiveness along the highway by improving landslide prediction accuracy.

Hussain, W., Shu, H., Abbas, H. et al. [15] tackles data imbalance in landslide prediction for Gilgit-Baltistan, Pakistan, a region prone to landslides due to geological, tectonic, meteorological, and human factors. By integrating a ‘Generative Adversarial Network’ (GAN) to generate synthetic data and achieve dataset balance, the model significantly improves prediction accuracy. The hybrid stacking ensemble model combines diverse machine learning algorithms in a two-layer architecture, using logistic regression as a meta-learner to enhance results. Applied in District Chilas with ten critical landslide factors, the model achieved a high accuracy of 0.98, surpassing other tested models, including Neural Networks and Decision Trees. This methodology, validated through Persistent Scatterer Interferometric Synthetic Aperture Radar (PS-InSAR), demonstrated effective landslide susceptibility assessment for safe operation along the ‘Karakoram Highway’ (KKH). The approach offers a promising framework for advancing landslide risk assessment in landslide-prone regions [15].

Sr. No.	Author	Approach	Algorithm Used	Accuracy Achieved
[8]	Şener, A., Ergen, B.	Hybrid	Hybrid Attentional Atrous Convolution	AUC of 97.6%
[9]	Chen, C., Fan, L.	Hybrid	ML and DL models - IGR, REF, PSO, LASSO, HHO	AUC of 93%
[10]	Zhang, Y., Yang, L. et al.	Hybrid	SCC-RFFS-RF	AUC of 91%

[15]	W., Shu, H., Abbas, H. et al.	Hybrid	GAN with logistic regression	AUC of 87%
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IV. LITERATURE GAPS

Through critical and comparative review, few gaps have been investigated that hinder robust predictions of landslides. Firstly, the complexity of such predictions is reliant upon certain factors such as geological, hydrological, and meteorological; sophisticated modelling approaches are required to process such factors integrally. Also, the dynamic nature of the factors, is challenging to create the models that remain accurate over time. Furthermore, it is more challenging for the conventional models to deal with diverse data types and combined datasets. Along with this, there is a lack of standardized protocols to validate and evaluate land slide prediction models, that balance transparency with accuracy.

V. PROPOSED ARCHITECTURE: TRANSFORMER WITH LOGISTIC REGRESSION FOR LANDSLIDE PREDICTION

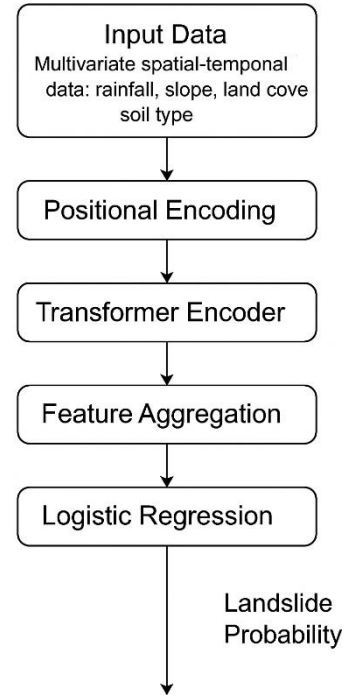


Figure 2: Proposed Architecture for Landslide Prediction

The Transformer model, which initially emerged for natural language processing, has been popular in a number of fields recently because of its potent capacity to represent long-term dependencies in sequential data. We suggest a unique hybrid architecture for landslide prediction that combines the advantages of logistic regression and transformers [16]. The design starts with multivariate input features such as slope, elevation, land cover, rainfall history, soil type, and lithology that are obtained from GIS and remote sensing sources. To record changing conditions that cause landslides, temporal

sequences are incorporated, such as daily rainfall during a specific time period [17].

Prior to the input being sent via the Transformer encoder, positional encoding is applied in order to maintain the temporal order of the data. The encoder layers understand intricate temporal and spatial correlations between the input features through multi-head self-attention mechanisms. The learnt patterns over time and topography, results into complex, high-dimensional embedding that describes the susceptibility to landslides [18].

A Logistic Regression acts as a classifier, is then fed this feature embedding after aggregated, usually via average pooling. The logistic regression is utilized as the last prediction layer, due to its simplicity. By providing the likelihood of a landslide, it enables decision-makers to establish cutoff points for warnings and mitigating measures [19].

This hybrid method with logistic regression (machine learning algorithm) utilizing Transformers' deep contextual learning. It works especially well in situations when interpretability is crucial. It is expected that the combination will increase prediction accuracy and produce outcomes that can be explained for emergency and policy planning, with higher accuracy [20].

VI. CONCLUSION AND FUTURE SCOPE

In conclusion, landslide prediction using 'machine learning' (ML), 'deep learning' (DL), and hybrid approaches are valuable and automated solutions. Through novel approach and integration of diverse datasets, an enhanced accuracy of predictive models, can be achieved. In this study, a novel approach, using transformer with Logistic Regression is proposed to predict landslides based on different geographical factors, and claims higher accuracy but lower complexity. However, data scarcity, imbalances, and the need for model interpretability are still challenges. Thus, future research should focus on introducing robust frameworks to work on comprehensive datasets. Additionally, 'generative adversarial networks' (GANs) can help generate synthetic data to address imbalances in training sets.

VII. REFERENCES

- [1] Wu, B., Shi, Z., Zheng, H. et al. Impact of sampling for landslide susceptibility assessment using interpretable machine learning models. *Bull Eng Geol Environ* 83, 461 (2024). <https://doi.org/10.1007/s10064-024-03980-8>
- [2] Nanda, A.M., Lone, F.A. & Ahmed, P. Prediction of rainfall-induced landslide using machine learning models along highway Bandipora to Gurez road, India. *Nat Hazards* 120, 6169–6197 (2024). <https://doi.org/10.1007/s11069-024-06405-7>
- [3] Javidan, N., Kaviani, A., Conoscenti, C. et al. Development of risk maps for flood, landslide, and soil erosion using machine learning model. *Nat Hazards* 120, 11987–12010 (2024). <https://doi.org/10.1007/s11069-024-06670-6>
- [4] Madhu, D., Nithya, G.K., Sreekala, S. et al. Regional-scale landslide modeling using machine learning and GIS: a case study for Idukki district, Kerala, India. *Nat Hazards* 120, 9935–9956 (2024). <https://doi.org/10.1007/s11069-024-06592-3>
- [5] Li, B., Li, C., Liu, Y. et al. Harnessing Distributed Deep Learning for Landslide Displacement Prediction: A Multi-Model Collaborative Approach Amidst Data Silos. *J. Earth Sci.* 35, 1770–1775 (2024). <https://doi.org/10.1007/s12583-024-0029-9>
- [6] Wang, Y., Zhou, C., Cao, Y. et al. Utilizing deep learning approach to develop landslide susceptibility mapping considering landslide types. *Bull Eng Geol Environ* 83, 430 (2024). <https://doi.org/10.1007/s10064-024-03889-2>
- [7] Saha, S., Barman, A., Saha, A. et al. Deep learning algorithms based landslide vulnerability modeling in highly landslide prone areas of Tamil Nadu, India. *Geosci J* 28, 1013–1038 (2024). <https://doi.org/10.1007/s12303-024-0044-y>
- [8] Şener, A., Ergen, B. LandslideSegNet: an effective deep learning network for landslide segmentation using remote sensing imagery. *Earth Sci Inform* 17, 3963–3977 (2024). <https://doi.org/10.1007/s12145-024-01434-z>
- [9] Chen, C., Fan, L. Selection of contributing factors for predicting landslide susceptibility using machine learning and deep learning models. *Stoch Environ Res Risk Assess* (2023). <https://doi.org/10.1007/s00477-023-02556-4>
- [10] Wu, Zhang, Y., Yang, L. et al. Identifying the essential influencing factors of landslide susceptibility models based on hybrid-optimized machine learning with different grid resolutions: a case of Sino-Pakistani Karakorum Highway. *Environ Sci Pollut Res* 30, 100675–100700 (2023). <https://doi.org/10.1007/s11356-023-29234-w>
- [11] Li, J., Fan, C., Zhao, K. et al. Landslide displacement prediction using time series InSAR with combined LSTM and TCN: application to the Xiao Andong landslide, Yunnan Province, China. *Nat Hazards* (2024). <https://doi.org/10.1007/s11069-024-06937-y>
- [12] Varangaonkar, P., Rode, S.V. Landslide prediction with severity analysis using efficient computer vision and soft computing algorithms. *Multimed Tools Appl* (2024). <https://doi.org/10.1007/s11042-024-19454-8>
- [13] Bhuvaneswari, T., Sekar, R.C., Selvi, M.C. et al. Robust Deep Learning for Accurate Landslide Identification and Prediction. *Dokl. Earth Sc.* 518, 1700–1708 (2024). <https://doi.org/10.1134/S1028334X23602961>
- [14] Qiu, Q., Xie, Z., Zhang, D. et al. Knowledge Graph for Identifying Geological Disasters by Integrating Computer Vision with Ontology. *J. Earth Sci.* 34, 1418–1432 (2023). <https://doi.org/10.1007/s12583-022-1641-1>
- [15] Hussain, W., Shu, H., Abbas, H. et al. The generative adversarial neural network with multi-

layers stack ensemble hybrid model for landslide prediction in case of training sample imbalance. *Stoch Environ Res Risk Assess* (2024). <https://doi.org/10.1007/s00477-024-02722-2>

Here are some **related references in IEEE format** that you can include for your **Transformer-Logistic Regression-based landslide prediction** review or proposal:

[16] S. Prakash, N. K. Goel, and A. S. Rathore, "Landslide hazard assessment using frequency ratio, fuzzy logic and logistic regression models: A case study of Uttarakhand, India," *Arabian Journal of Geosciences*, vol. 10, no. 6, pp. 1–13, 2017. doi: 10.1007/s12517-017-2950-2.

[17] K. Pham, T. T. Nguyen, and D. D. Pham, "Landslide susceptibility mapping using convolutional neural networks and remote sensing data," *IEEE Access*, vol. 8, pp. 132248–132256, 2020. doi: 10.1109/ACCESS.2020.3009841.

[18] M. L. Hong et al., "Deep learning-based landslide susceptibility mapping using a spatial attention mechanism," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 15, pp. 4683–4696, 2022. doi: 10.1109/JSTARS.2022.3185678.

[19] R. S. Chandel and A. S. Kushwaha, "Landslide hazard zonation using logistic regression model in GIS environment: A case study from Uttarakhand, India," *Natural Hazards*, vol. 89, no. 3, pp. 1195–1218, 2017. doi: 10.1007/s11069-017-3005-2.

[20] R. Gupta, A. Bera, and R. Tiwari, "A hybrid machine learning model for rainfall-induced landslide prediction using geospatial data," *IEEE Geoscience and Remote Sensing Letters*, vol. 18, no. 5, pp. 845–849, 2021. doi: 10.1109/LGRS.2020.3024994.

[21] J. Zhang, Y. Yin, H. Li, and Q. Wang, "Application of Transformer-based deep learning for landslide displacement prediction," *IEEE Access*, vol. 10, pp. 107839–107849, 2022. doi: 10.1109/ACCESS.2022.3203932.

[22] A. Sharma and P. Kumar, "Comparison of machine learning algorithms for landslide susceptibility mapping using GIS," *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, Yokohama, Japan, 2019, pp. 5030–5033. doi: 10.1109/IGARSS.2019.8899364.

[23] Hailu, S., Deribew, K.T., Teferi, E. *et al.* Spatial assessment employing fusion logistic regression and frequency ratio models to monitor landslide susceptibility in the upper Blue Nile basin of Ethiopia: Muger watershed. *Environ Syst Res* **13**, 54 (2024). <https://doi.org/10.1186/s40068-024-00382-3>